

NANYANG JUNIOR COLLEGE
JC 2 PRELIMINARY EXAMINATIONS
Higher 2

CANDIDATE
NAME

CLASS

BIOLOGY

Paper 3 Applications and Planning

Additional Materials: Answer Paper

9648/03

Sept 2014

2 hours

READ THESE INSTRUCTIONS FIRST

Write your name and CT on all the work you hand in.
Write in dark blue or black pen on both sides of the paper.
You may use soft pencil for any diagrams, graphs or rough working.
Do not use staples, paper clips, highlighters, glue or correction fluid.

Answer **all** questions.

At the end of the examination, fasten all your work securely together.
The number of marks is given in brackets [] at the end of each question or part question.

Answer questions **1** to **4** in the spaces provided on the question paper.

Answer question **5** on the separate answer paper provided.

For Examiner's Use	
1	15
2	13
3	12
5	20
Total	60
For Examiner's Use	
4	12

This document consists of 15 printed pages and 1 blank page.

[Turn over

Section A

Answer **all** the questions in this section.

- 1** Restriction enzymes cut DNA molecules only at specific target sites with particular base sequences. The target sites for the enzymes *EcoRI*, *BamHI* and *HpaI* are shown in **Table 1.1**. The lines drawn in each sequence show where the enzyme cuts the DNA molecule.

Table 1.1

restriction enzyme	specific target base sequence of DNA
<i>EcoRI</i>	<pre> G A A T T C C T T A A G </pre>
<i>BamHI</i>	<pre> G G A T C C C C T A G G </pre>
<i>HpaI</i>	<pre> G T T A A C C A A T T G </pre>

A bacterial plasmid includes the base sequence shown in **Fig. 1.1**. There are no target sequences for these three restriction enzymes in the rest of the plasmid.

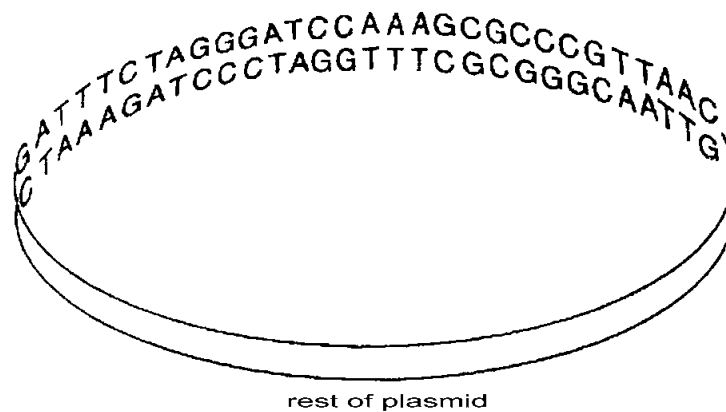


Fig. 1.1

- (a)** State how many restriction fragments of DNA will be present after the plasmid has been treated with.

(i) *HpaI* and *EcoRI*;

1

[1]

(ii) *HpaI* and *BamHI*;

2

[1]

- (b) A genetic engineer cuts the plasmid shown in **Fig. 1.1** with *HpaI* in order to insert a human gene.

- (i) Describe the steps that must be subsequently followed to produce a recombinant plasmid.

DNA fragments and plasmid cut by *HpaI* possesses blunt ends; (not a marking point)

1. Use terminal transferases to add nucleotides to one strand of DNA to form complementary sticky ends on cut DNA fragments and cut plasmid/ add linker DNA and then cut with another RE to create sticky ends;
2. complementary bases of DNA fragment and plasmid base pair to form hydrogen bonds;
3. DNA ligase covalently join plasmid, and target DNA together to form recombinant plasmid via phosphodiester bonds;

[3]

- (ii) Describe any difference in the procedure that would be needed to produce a recombinant plasmid if the plasmid were cut with *BamHI* instead of *HpaI*.

DNA fragments cut by *BamHI* possesses sticky ends: 'GATC' while DNA fragments cut by *HpaI* possesses blunt ends;

enzyme terminal transferases do not need to be added to fragments cut by *BamHI*; additional nucleotides for sticky ends formation is not added to DNA fragments and plasmid cut by *BamHI*;

DNA fragment also has to be cut with *BamHI* to produce complementary sticky ends;

[2]

A length of DNA from one of a pair of homologous chromosomes is shown in **Fig. 1.2**. The target sites of *EcoRI* are shown by arrows and the length of DNA between the target sites is given in kilobases (kb).

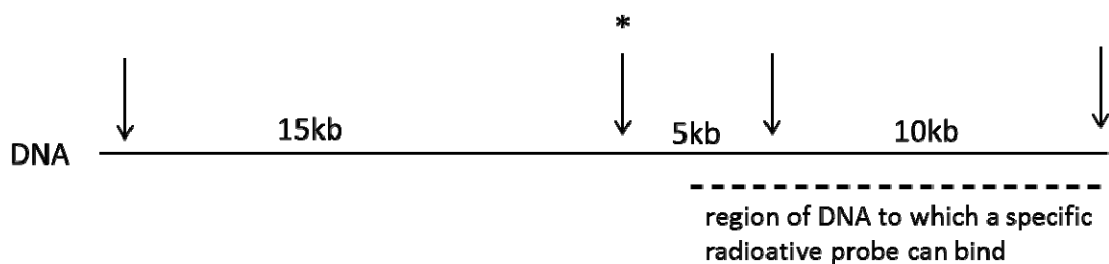
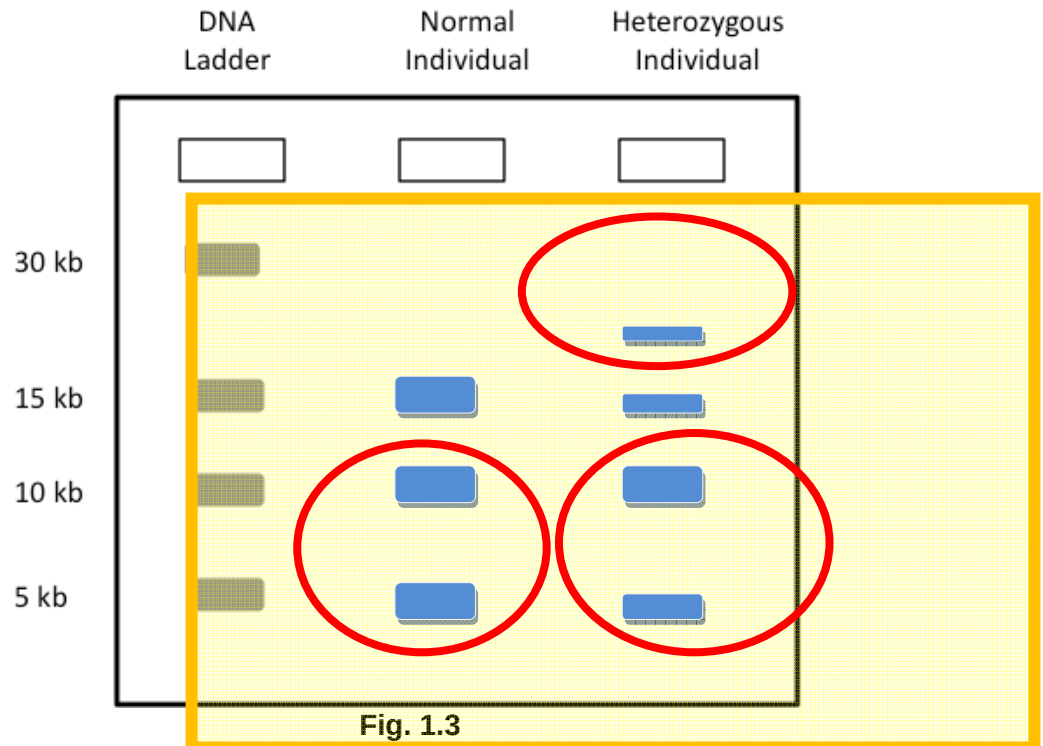


Fig. 1.2

A mutation alters one base of the coding of the site marked with an asterisk (*). DNA from an individual who is heterozygous for this mutation is cut with *EcoRI* and the DNA fragments separated by electrophoresis.

(c) On Fig. 2.3,

- (i) Draw all the fragments of this length of DNA produced from the pair of homologous chromosomes in each of the individuals tested and label the size of the fragments produced.



1m – normal individual (3 bands)

1m – heterozygous individual (4 bands)

1m – difference in thickness of bands

[3]

- (ii) Circle the fragments which will combine with the radioactive probe and appear as black bands on an autoradiography;

1m – 10kb and 5 kb on normal individual

1m – 20kb, 10kb, 5kb on heterozygous individual

[2]

- (d) DNA fragments are separated by electrophoresis in genetic fingerprinting. Briefly describe how DNA fragments may be separated by electrophoresis.

DNA samples loaded into wells at negative end of agarose gel (molecular sieve)/
Negative electrode at end where DNA is loaded as DNA is negative;

DNA samples migrate from negative end to positive end when electric current is applied;

Agarose gel acts as a molecular sieve;

Rate of movement inversely proportionally to size of DNA fragments/larger fragments move slower/ DNA separated in terms of their size with smaller fragments furthest from wells; (compulsory point)

[3]

[Total: 15]

- 2 The first gene therapy trial for an inherited disorder was initiated on 14 September 1990. The trial was for a patient named Ashanthi DeSilva, who suffered from a very rare recessively inherited disorder, adenosine deaminase (ADA) deficiency.

An inherited deficiency of ADA has particularly severe consequences in the case of T lymphocytes. As a result, ADA-deficient patients suffer from severe combined immunodeficiency (SCID).

- (a) Explain how a mutation of ADA gene affects the immune system of ADA-deficient SCID patients.

ADA-deficient SCID patients are homozygous for mutated ADA allele on chromosome 20, the mutant alleles do not code for the functional ADA / the normal alleles code for functional ADA;

The normal function of ADA is to catabolise purines / breakdown of deoxyadenosine/ purine metabolism, hence the absence of functional ADA causes the accumulation/high levels of purines / deoxyadenosine in cells;

High level of purines are toxic to T cells & B cells, hence prevents the development of T and B cells / does not allow lymphocyte proliferation;

.....
.....
.....

- (b) Suggest why a recessively inherited disorder such as ADA-SCID can be treated by gene therapy. [3]

Ref to single gene disorder / loss-of-function mutation;

Introduction of one normal / dominant allele to produce sufficient amounts of functional gene product / ADA to mask effect of recessive allele;

If it is a dominant disorder, both mutant alleles must be replaced/inactivated;

.....
.....
..... [2]

- (c) The novel ADA gene therapy approach involved essentially the following steps:

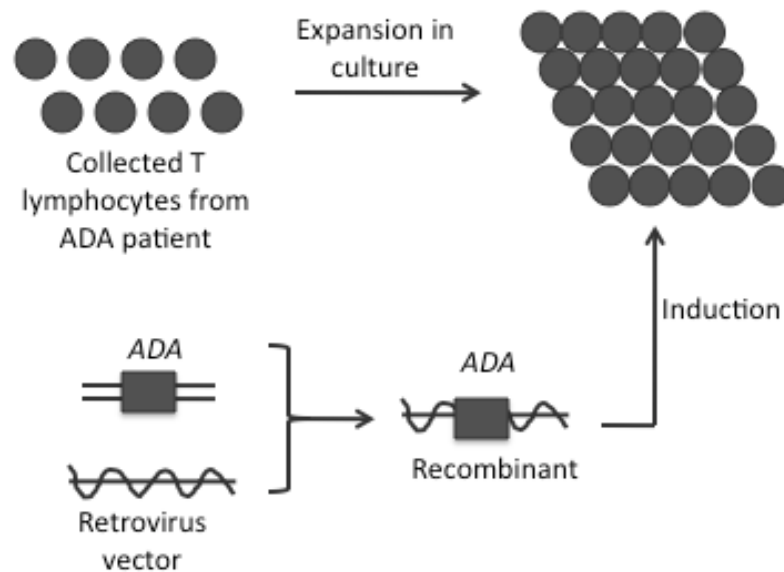


Fig. 2.1

- (i) Suggest why the *ex vivo* method was used in this case.

T-lymphocytes can be easily transfected and cultured in the laboratory to form larger numbers;
 Allow for selection of successfully transformed cells;
 Prevent the retrovirus from targeting other rapidly proliferating cells;

- (ii) Suggest an improvement to the procedure in **Fig. 2.1** so as to increase the success rate of the treatment. [1]

- culture blood stem cells / hematopoietic stem cells instead of T-lymphocytes taken from the bone marrow / umbilical cord;
- differentiate to form all types of **white blood cells**, including B- and T-lymphocytes
- since stem cells are able to self-renew by mitosis for a long period of time
- may provide more **long term cure / no need for repeated treatment**

OR

- **select for / identify** successfully transformed cells after **infection of cells with retrovirus**
- **increase transformed cells** through cell culture
- before **re-implanting cells into patient** ensure that **re-introduced cells** are **all able to express normal ADA gene**

- (d) Gene therapy using the same form of treatment as the above case study has succeeded for 15 French baby boys suffering from X-linked SCID. However, three of the little boys later developed cancer and died. [3]

Suggest how this is possible.

inability to control the specific site of insertion / random integration of transgene / insertional mutagenesis;

disrupt gene control regions / elements + (example): resulting in activation of proto-oncogenes (becoming oncogenes) due to mutation of one allele/one copy
OR
inactivation of two alleles / two copies of the tumour suppressor gene;

Tumour suppressor gene is a gene involved in promoting apoptosis / inhibiting cell division / POG is involved in promoting cell division, which when mutated, which when mutated, results in uncontrolled cell division/proliferation

Reject: statements that imply insertional inactivation of proto-oncogenes contributes to cancer.

[2]

(e) State and explain another factor that hinders the effectiveness of current gene therapy approaches.

1. Gene therapy is short-lived and requires frequent / repeated treatments;
 - a. **target cells** may be **short-lived** or may become unstable /short lived after introduction of therapeutic DNA
 - b. difficult to ensure DNA is integrated into target cell genome. **rapidly dividing nature** of cells prevents efficiency of treatment if gene is extrachromosomal
 - c. result: patient often needs to **undergo many rounds of gene therapy** for a long / indefinite period of time
 1. Difficult to treat genetic diseases caused by combined effects of many defective genes / multi-gene disease;
 - a. Difficult to introduce many genes at the same time ; require **multiple replacement of defective alleles**
 2. Immune response may be triggered as a foreign viral vector is introduced / virus may regain / develop virulence and cause disease;
 3. Problem with controlling / regulating the activity of gene expression;
 - a. ref. cells make **appropriate amounts** of gene product / low levels of expression
 - b. inserted genes may not be expressed (if integrated into a **highly condensed heterochromatic region**)
 - c. at the **right time** and in the **right location** of the body
 - d. ref. difficulty controlling if the transgene is **correctly integrated into the genome**
 - e. development of cancer due to insertional inactivation of control elements / tumour suppressor gene or proto-oncogene (becoming oncogene in the process), which affect gene expression
 4. Inability to target multiple diseased sites;
 5. Problem with dealing with dominant condition / non dividing cells ;
- Reference to problems with liposomes (low toxicity but low efficiency)
-

[2]
[Total: 13]

- 3** Deficiency of vitamin A in the diet is a major problem in the rice-based societies of South-East Asia. Rice plants produce the precursor of vitamin A, β -carotene, in their green tissues, but not in the edible endosperms of their seeds.

Rice seeds lack the enzymes for two steps of the pathway for β -carotene production. The genes for these two enzymes have been inserted into rice embryos by genetic engineering, giving rise to Golden Rice™. This rice produces seeds containing β -carotene.

The added genes were:

- the *psy* gene from daffodil plants,
- the *crt 1* gene from the bacterium *Erwinia uredovora*.

- (a)** Explain why it is possible for rice plants to express genes from a bacterium.
 All organisms share the same genetic code;
 eukaryotic promoter from rice plant can be inserted upstream of bacterial genes;

[1]

The following steps were taken to produce Golden Rice™:

- **Step 1:**
A length of DNA that included a rice endosperm-specific promoter, *psy*, *crt 1* was produced.
- **Step 2:**
Copies of this length of DNA were inserted into plasmids from the bacterium *Agrobacterium tumefaciens*.
- **Step 3:**
A. tumefaciens containing the plasmids were mixed with rice embryos in tissue culture
- **Step 4:**
The embryos were grown into plantlets and then plants.

- (b)** Describe the role of the rice endosperm-specific promoter that was added to *psy* and *crt 1* in **step 1**.

Role of promoter is to regulate expression of *psy* and *crt 1* genes;
 Promoter is specific to rice endosperm as transcription factors that can bind to the promoter are found only in the endosperm;
 This ensures that transcription will only take place in the rice endosperm / edible part of seed;

[2]

The concentration of β -carotene was increased by further genetic engineering. More recent strains, such as Golden Rice 2™, contain about $31 \mu\text{g g}^{-1}$ of β -carotene in the endosperm of their seeds.

- (c)** Suggest what further genetic engineering was carried out to increase the concentration of β -carotene in rice seeds.

1. Increase expression of *psy* / *crt 1* gene using genes from other sources;;
2. by inserting the DNA length downstream of an active promoter;;
3. To introduce more than one copy of gene into rice embryos;;

-
-
-
- [1]
- (d) Field trials of genetically engineered crops, such as Golden Rice™, hope to identify any risks to the environment or to human health of growing and eating the crops. Suggest one possible risks to the environment of growing genetically engineered crops;

Gene transfer to other plants/crops due to cross pollination;

Contamination of other crop (e.g. ref to organic crop);

Gene transfer to wild relative;

Gene transfer from crop to crop via bacteria or viruses with unknown effect;;

Recipient plant (plant that receives the gene) outcompetes others / ref to 'superweed';

Toxic to non pests;

.....

.....

[1]

In another variety of rice plant, an enzyme, acetolactate synthase (ALS) catalyse the first step in the biosynthesis of valine and isoleucine. A common herbicide used by farmers, chlorsulfuron, works by binding to ALS. Thus, a single amino acid change in the ALS protein confers the plant resistance to the herbicide chlorsulfuron.

In some rice plants, the allele that results in abnormal ALS protein was identified. The presence of this allele makes the rice plant naturally resistant to the herbicide.

This allele was inserted into a vector and introduced into *Agrobacterium tumefaciens*. Rice protoplasts were transformed with *A.tumefaciens* carrying this vector and the transformed protoplasts could be stimulated to divide by mitosis to form calli and regenerated into whole plants by tissue culture techniques.

- (e) With reference to tissue culture techniques, describe how plant growth regulators can result in plantlet formation from calli.

Low cytokinin to high auxin ratio triggers root growth;;

High cytokinin to auxin ratio triggers shoot growth;;

.....

.....

[2]

The effectiveness of abnormal ALS protein was tested using two groups of rice plants. One group was given fertilizer that contains valine and isoleucine while the other group was given fertilizer which does not contain valine and isoleucine.

Each group of rice plants consists of plant type X, Y and Z and both groups are subjected to identical growing conditions and to herbicide spraying at regular intervals.

- X: Susceptible rice plants genetically engineered, as described above to express the allele coding for abnormal ALS protein

Y: Susceptible rice plants

Z: Resistant rice plants

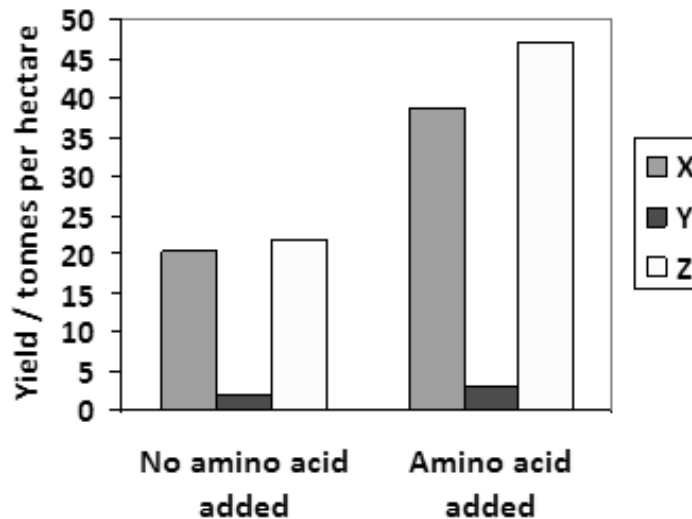


Fig. 3.1

- (f) With reference to Fig. 3.1,
 (i) describe and explain the effects of herbicides on yield of plant type X and Z when amino acids are added.

Yield for resistant rice plants is slightly more than genetically engineered rice plants (X), between 1 – 8 tonnes per hectare;

as some of the protoplasts are not transformed with the abnormal ALS protein allele, therefore resistance to herbicide is not conferred;

[2]

- (ii) explain the effects of amino acids to plant types X and Z.

the addition of amino acids increased the yield of the GE (X) and resistant rice plants (Z) compared to when no amino acids were added;

valine and isoleucine are essential amino acids for plants to synthesise proteins;

addition of valine and isoleucine allows the plant to continue with the biosynthesis without the presence of a functional protein / (alleviates herbicide-induced growth inhibition, as these amino acids are essential for synthesis of proteins e.g. proteins found on cell membrane in the rice plants;)

[3]

[Total: 12]

4 Planning Question

Electroporation is a method that is used to introduce foreign DNA into *E. coli* cells. This involves subjecting a mixture of bacteria and plasmid to a brief electric pulse.

You are required to plan, but not carry out, an investigation into the effect of voltage on the efficiency of transformation by electroporation.

In this investigation, pBR322 (**Fig. 4.1**) was used to transform the *E. coli* cells. pBR322 contains both ampicillin and tetracycline resistance genes.

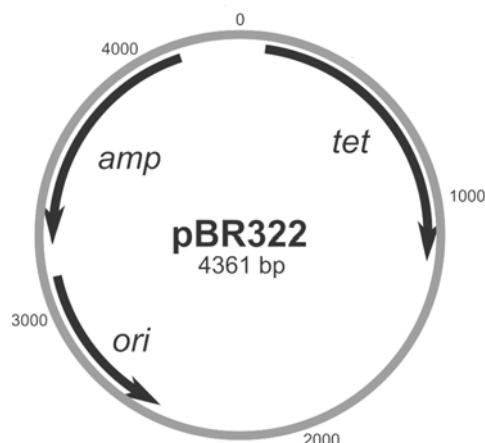


Fig. 4.1

An electroporator is used to carry out electroporation. The voltage and length of the electrical pulse can be varied with this machine. The range of voltage used is typically 1400 – 2000 V and the length of electrical impulse between 4-6 ms. A voltage which is too high will cause irreversible changes in membrane permeability. The mixture of bacteria and plasmid, which must be kept on ice prior to electroporation, is put into a cuvette which is then placed into a chamber in the electroporator (**Fig. 4.2**).

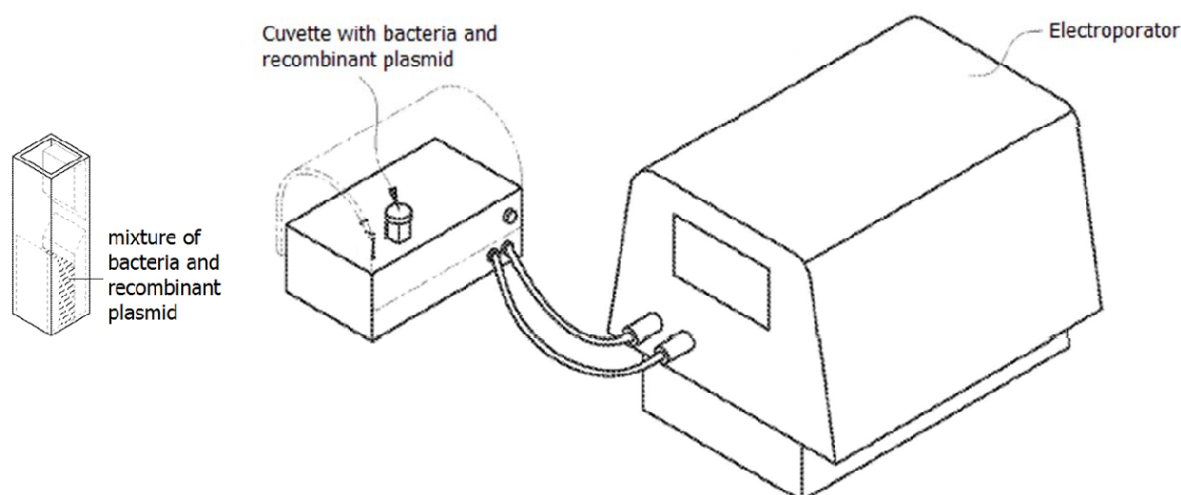


Fig. 4.2

After electroporation, the mixture of bacteria must be quickly resuspended in SOC broth (a type of growth medium) and incubated at 37°C for 1 hour for the surviving cells to recover. This is then followed by plating the bacteria on agar and incubation at 37°C for 16-24 hours. The plates can then be examined to calculate transformation efficiency.

Transformation efficiency can be calculated using the following equation:

$$\text{Transformation efficiency / cfu } \mu\text{g}^{-1} = \frac{\text{Total number of colonies on agar plate}}{\text{Amount of DNA plated (in } \mu\text{g)}}$$

*cfu: colony forming unit

Your planning must be based on the assumption that you have been provided with the following equipment and materials which you must use:

- 100 μl *E. coli* cells in electroporation buffer
- 1 μl recombinant DNA (concentration of 0.1 $\mu\text{g} / \mu\text{l}$)
- 1 μl sterile water
- Ice bucket with ice
- Electroporator
- Sterile cuvettes for electroporation (max. volume of 150 μl)
- Agar plates with ampicillin
- Sterile spreader for spreading bacterial culture on agar plate
- SOC broth
- Incubator
- A variety of micropipettes, sterile micropipette tips and sterile 2 ml microcentrifuge tubes

Your plan should have a clear and helpful structure to include

- an explanation of the theory to support your practical procedure
- a description of the method used, including the scientific reasoning behind the method, a control experiment and any recommended safety measures
- an explanation of the dependent and independent variables involved
- relevant, clearly labelled diagram/s showing selection of the recombinant bacteria on plates
- how you will record your results and ensure that they are as accurate and reliable as possible
- proposed layout of results tables and graphs with clear headings and labels
- the correct use of technical and scientific terms

[Total:12]

Theoretical background

- As voltage increases, transformation efficiency increases up to a point where membrane is irreversibly damaged, leading a drop in transformation efficiency as voltage further increases /

- At voltages which are too high or too low, the bacteria's ability to take up DNA by transformation becomes reduced, and at high voltages, many/most bacteria will die.
- Only transformed bacteria will be able to grow/form colonies on the agar plates with ampicillin. These colonies should also fluoresce under UV light.

Dependent variable: number of colonies that grow on agar plate (i.e. transformation efficiency)

Independent variable: voltage (at least 5 different voltage: between 1400 – 2000v)

Constant variables: volume of *E. coli* cells (e.g. 20µl), volume of recombinant DNA (e.g. 0.2µl), length of electrical pulse

Hypothesis: Transformation efficiency decreases with increase in voltage past an optimum.

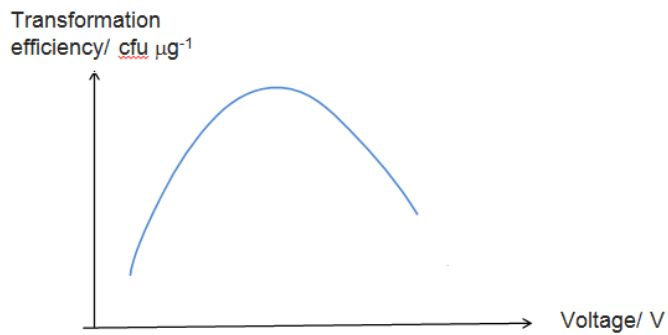
Procedures

Recommended Procedure:

1. Label 5 electroporation cuvettes with the 5 different voltages to be used (i.e. between 1400-2000v).
2. Using aseptic techniques, transfer 20µl of cells and 0.2µl of cold DNA into each of the electroporation cuvettes. Keep the mixture on ice for 10 minutes.
3. Place the cuvette labelled 1400v into the electroporator and for 5ms.
4. Immediately following electroporation, resuspend the bacterial mixture in 1ml of SOC broth, in a microfuge tube. Incubate at 37°C for 1 hour.
5. Repeat steps 3 to 5 for the cuvettes labelled with the other 4 voltages stated in the table.
6. Label the edge 5 petri dishes with the respective 5 voltages.
7. Using sterile spreaders, spread bacterial culture onto the respectively labelled petri dishes.
8. Tape the petri dishes with parafilm and invert them to be incubated at 37°C 16 – 24h.
9. Count the number of bacterial colonies on each plate and record the data in a table.
10. Repeat procedure twice to obtain a set of 3 results to obtain a mean.

Data Presentation

Tube	Voltage/ V	Number of colonies				Transformation efficiency/ cfu µg ⁻¹
		Plate 1	Plate 2	Plate 3	mean	
	1400					
	1550					
	1700					
	1850					
	2000					



Risk assessment

1. Wear gloves when handling *E. coli* to avoid contact with microorganisms
2. Wipe table with antiseptic solution (such as ethanol/chlorox) to prevent growth of harmful microorganisms
3. Proper disposal/ treatment of contaminated materials or equipment using steriliser/ autoclave to prevent growth of microorganisms.
4. Handle the electroporator with dry hands to prevent electrocution
5. Use the UV shields to cut down on exposure to UV light

Mark Scheme:

- 1 electroporation creates/ allows transient pores/ gaps to occur in the cell membrane
- 2 As voltage increases, transformation efficiency increases up to a point where membrane is irreversibly damaged, leading a drop in transformation efficiency as voltage further increases /At voltages which are too high or too low, the bacteria's ability to take up DNA by transformation becomes reduced, and at high voltages, many/most bacteria will die.
- 3 Only transformed bacteria will be able to grow/form colonies on the agar plates with ampicillin due to resistance conferred by the antibiotic resistance genes
- 4 At least 5 different voltages (between 1400 to 2000 V), regular spacing between intervals
- 5 (a) Constant volume of *E. coli* cells (eg. 20 μ l) and
(b) constant volume of recombinant DNA (eg. 0.2 μ l)
- 6 Specify time for incubation of bacteria and recombinant plasmid in ice (1 - 20 minutes). @ incubate together or separately.
- 7 Specify fixed time for electroporation (4 – 6 ms)
- 8 Specify fixed vol of SOC (>2ml) and incubate at 37°C for 1 hr
- 9 transformation mixture plated onto agar plates using spreading method
- 10 (a) Incubate the plates at 37°C
(b) for fixed no. of hours between 16-24hr @overnight
- 11 Control experiment – (a) *E. coli* cells + sterile water subjected to electroporation (no colonies observed)
(b) subjected to same conditions as mixtures with plasmid / repeat exp/ OWTTE
- 12 Count number of colonies which grow on plates
- 13 (a) Replicates
(b) At least two repeats for calculation of mean
- 14 Tabulation:
(a) Correct headings: colony count, mean, voltage, average transformation efficiency/cfu μ g⁻¹
- 15 Plot graph (with trend line) of ave. transformation efficiency/ cfu μ g⁻¹ against voltage
- 16 Drawing of Petri + appropriate labelling
- 17 Two safety precautions / risk hazards identified
(a) Wear gloves when handling *E. coli* to avoid contact with microorganisms
(b) Wipe table with antiseptic solution (such as ethanol/chlorox) to prevent growth of harmful microorganisms
(c) Proper disposal/treatment of contaminated materials or equipment using sterilizer/ autoclave to prevent growth of harmful microorganisms
(d) Handle (plug of) electroporator with dry hands to prevent electrocution.
- 18 The correct use of technical and scientific terms
(a) transient pores/ gaps
(b) ref. to transformed cells being resistant to ampicillin/being able to grow on ampicillin

1 – 3: Max 2m

4 – 16: Max 10m

17 – 2 precautions for 1m

(c) 18 – max 1

5 Free-response question

Write your answers on the separate answer paper provided.

Your answers:

- should be illustrated by large, clearly labelled diagrams, where appropriate.
- must be in continuous prose, where appropriate.
- must be set out in sections **(a)**, **(b)** etc., as indicated in the question.

- (a) Describe the polymerase chain reaction (PCR) and explain the limitations of the procedure. [8]
- (b) Discuss the benefits and ethical concerns of the human genome project for humans. [6]
- (c) Discuss the social implications of genetically modified crop plants. [6]

[Total: 20]

Describe the polymerase chain reaction (PCR) and explain the limitations of the procedure. [8]

Description of PCR

- 1) PCR is a process used to, amplify DNA / synthesize large amounts of DNA from a minute amount of starting material;
- 2) Thermostable, DNA polymerase / *Taq* polymerase is used;
- 3) Two DNA primers / single-stranded DNA molecules which are complementary to the flanking sequence / start and end of the segment to be amplified;
- 4) Denaturation: heating it to 95°C + involves denaturation of double-stranded DNA into single-stranded DNA / breaking of hydrogen bonds between nitrogenous bases of the two strands;
- 5) Annealing: temperature lowered to 50 – 60°C + allows primers to anneal via complementary base pairing due to hydrogen bonding;
- 6) Elongation: temperature increased to 72°C + *Taq* polymerase adds nucleotides to free 3'-OH end of primers using the DNA molecule as a template;
- 7) Sequential process of denaturation-annealing-elongation is repeated many times / products of the previous reaction are used as reactants in the next cycle;

Limitations of PCR (max 3 marks)

8. Knowledge of the DNA or amino acid sequence of desired gene or protein is needed to synthesise flanking nucleotide primers;
9. Non-target DNA sequence may be amplified instead of the desired sequence as primers are short nucleotide sequences and may not be specific enough;
10. *Taq* polymerase, of bacteria origin, does not perform proofreading, thus, there may be mistakes in complementarity of the nucleotides added;
11. Highly sensitive to contamination, thus non-target sequences can be amplified;

Discuss the benefits and difficult ethical concerns of the human genome project for humans. [6] (H2)

Benefits (Max 4): State + describe one example within category

Molecular Medicine:

Improved diagnosis of disease/

Earlier detection of genetic predispositions to disease/
 Rational drug design/
 Gene therapy and control systems for drugs/
 Pharmacogenomics "custom drugs";

DNA Forensics (Identification)

Identify potential suspects whose DNA may match evidence left at crime scenes;
 Exonerate persons wrongly accused of crimes/
 Identify crime and catastrophe victims/
 Establish paternity and other family relationships/
 Identify endangered and protected species as an aid to wildlife officials (could be used for prosecuting poachers)/
 Detect bacteria and other organisms that may pollute air, water, soil, and food;
 Match organ donors with recipients in transplant programs/
 Determine pedigree for seed or livestock breeds/
 Authenticate consumables such as caviar and wine;

Risk Assessment

Assess health damage and risks caused by radiation exposure, including low-dose exposures/
 Assess health damage and risks caused by exposure to mutagenic chemicals and cancer-causing toxins/
 Reduce the likelihood of heritable mutations;

Bioarchaeology, Anthropology, Evolution, and Human Migration

Study evolution through germline mutations in lineages/
 Study migration of different population groups based on female genetic inheritance/
 Study mutations on the Y chromosome to trace lineage and migration of males/
 Compare breakpoints in the evolution of mutations with ages of populations and historical events

Energy and Environmental Applications

Use microbial genomics research to create new energy sources (biofuels)/
 Use microbial genomics research to develop environmental monitoring techniques to detect pollutants/
 Use microbial genomics research for safe, efficient environmental remediation/
 Use microbial genomics research for carbon sequestration;

Agriculture, Livestock Breeding, and Bioprocessing

Disease-, insect-, and drought-resistant crops;
 Healthier, more productive, disease-resistant farm animals;
 More nutritious produce;
 Biopesticides;
 Edible vaccines incorporated into food products;
 New environmental cleanup uses for plants like tobacco;

Ethical Concerns (Max 4)

Fairness in the use of genetic information by insurers, employers, courts, schools, adoption agencies, and the military, among others.

Privacy and confidentiality of genetic information;
 Psychological impact and stigmatization due to an individual's genetic differences;
 Reproductive issues including adequate informed consent for complex and potentially controversial procedures / use of genetic information in reproductive decision making / reproductive rights;
 Clinical issues including the education of doctors and other health service providers, patients, and the general public in genetic capabilities, scientific limitations, and social risks / implementation of standards and quality-control measures in testing procedures;
 Uncertainties associated with gene tests for susceptibilities and complex conditions (e.g., heart disease) linked to multiple genes and gene-environment interactions;
 Conceptual and philosophical implications regarding human responsibility, free-will versus genetic determinism / concepts of health and disease;
 Health and environmental issues concerning genetically modified foods (GM) and microbes;
 Commercialization of products including property rights (patents, copyrights, and trade secrets) and accessibility of data and materials;

Discuss the social implications of genetically modified crop plants. [6]

Positive

- Growth of pest-resistant plants means that lesser crops are lost due to insect damage / diseases spread by insect vectors, therefore resulting in higher yield.
- Lesser insecticide / pesticide is used, therefore the farmers can save more money.
- Growth of herbicide-resistant plants means that farmers can spray herbicides to kill weeds without harming the crops. This results in reduction in competition from weeds, resulting in higher crop yield.
- Savings for farmers translate into cheaper foods for consumers
- Enhanced quality of crop (e.g. Golden rice, expression of beta-carotene in endosperm in rice, which is converted into Vitamin A in the body when eaten) which helps to keep people healthy and to prevent malnutrition.

Negative

- There is a risk of transgene transfer to closely-related non-crop species. For e.g. for herbicide-resistant GM crops, the use of herbicides can lead to the rise of super-weeds that are resistant to herbicides as weeds are grown alongside herbicide-resistant GM crops, due to crossing with the closely related GM plant;
- Herbicide acts as a selection pressure when a grower continues to use only one particular herbicide without any other herbicide modes of action, or doesn't use any other cultural practices. The resistant weed type continues to survive, mature and produce seed. Subsequent populations of the resistant biotype will continue to increase, reducing crop yields.
- Farmers would then have to resort to heavy / excessive usage of herbicide, which persist in the environment (i.e. runoff of chemicals into waterways), affecting the ecosystem.
- Transfer of antibiotic resistance markers in GM crop to pathogenic microorganisms (gut) which may result in increase in resistance to clinically important antibiotics;
- Probability of introducing novel allergens as GM foods may contain proteins introduced from sources people are allergic to;
- Monopolistic behaviour of biotechnology companies as terminator gene is likely to be inserted into many GMO seeds, causing second generation seeds to be sterile;
- Reduction in biodiversity and changes in gene pool, when there is gene transfer of genetic material from GMO to other wild type / native population through e.g. pollination and dispersal.

- Increasing dependence of developing nations on industrialized nations as the GM crops / seeds may be too expensive for the poor famers in developing countries;